Device Information

CA3080

Printer Friendly Version

Operational Transconductance Amplifier

DS Datasheet & Related Docs | Description | Key Features | PT Parametric Data

Ordering Information

Part No.	Status	Temp.	Package	MSL	Price US $
CA3080E	InActive	Comm	8 Ld PDIP	N/A	
CA3080M	InActive	Comm	8 Ld SOIC	1	
CA3080M96	InActive	Comm	8 Ld SOIC	1	

The price listed is the manufacturer's suggested retail price for quantities between 100 and 999 units. However, prices in today's market are fluid and may change without notice.

MSL = Moisture Sensitivity Level - per IPC/JEDEC J-STD-020

SMD = Standard Microcircuit Drawing

Description

The CA3080 and CA3080A types are Gatable-Gain Blocks which utilize the unique operational-transconductanceamplifier (OTA) concept described in Application Note AN6668, “Applications of the CA3080 and CA3080A High- Performance Operational Transconductance Amplifiers”.

The CA3080 and CA3080A types have differential input and a single-ended, push-pull, class A output. In addition, these types have an amplifier bias input which may be used either for gating or for linear gain control. These types also have a high output impedance and their transconductance (gM) is directly proportional to the amplifier bias current (I_{ABC}).

The CA3080 and CA3080A types are notable for their excellent slew rate (50V/µs), which makes them especially useful for multiplexer and fast unity-gain voltage followers. These types are especially applicable for multiplexer applications because power is consumed only when the devices are in the “ON” channel state.

The CA3080A’s characteristics are specifically controlled for applications such as sample-hold, gain-control, multiplexing, etc.

Key Features

- Slew Rate (Unity Gain, Compensated) 50V/µs
- Adjustable Power Consumption 10µW to 30µW
- Flexible Supply Voltage Range ±2V to ±15V
- Fully Adjustable Gain 0 to gMRL Limit
- Tight gM Spread:
  - CA3080 2:1
  - CA3080A 1.6:1
- Extended gM Linearity 3 Decades

Related Documentation

Application Note(s):

- An IC Operational Transconductance Amplifier (OTA) With Power Capability
- Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers

Datasheet(s):

- Operational Transconductance Amplifier

Parametric Data

# of Amps	1
Slew Rate (V/µs)	75
V_S (min) (V)	±2
V_S (max) (V)	±15
BW @ -3dB (MHz)	2
Rail-to-Rail	N
Gain A_V (min) (V)	1
I_S(per amp) (mA)	1
I_{BIAS} (µA)	2000
V_{OS} (max) (mV)	.3
A_{VOL} or A_{ZOL} (dB or V/mA)	N/A
CMRR (dB)	110
PSRR (dB)	N/A

Applications

- Sample and Hold
- Multiplier
- Multiplexer
- Comparator
- Voltage Follower

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